Explanatory Note

Flex Fuel Vehicles

The Government has notified the regulation enabling the introduction of flex-fuel vehicles, designed to run on fuel between E20 [E(X) implies X% of ethanol in (100-X)% gasoline] and E85/E100. Flex-fuel technology is a proven and mature technology that has been deployed in countries such as Brazil and the United States.

Flex-fuel vehicles are specifically designed with fuel-system components and materials that are compatible with ethanol blends from E20 to E100. Vehicles designed as Flex fuel vehicles will not have material compatibility issues. Motorists should check the fuel compatibility sticker on their vehicle and follow the manufacturer's guidance in the owner's manual while opting for the fuel choice at the fuel station.

Advanced engine management systems automatically adjust vehicle operation according to the fuel composition, ensuring optimum performance under different driving conditions.

The introduction of flex-fuel vehicles in India is expected to generate employment opportunities through the development, manufacturing, and supply of flex-fuel vehicle components, ethanol production infrastructure, and related services.

Ethanol has a lower fossil carbon footprint than conventional gasoline. The use of higher ethanol blends can help reduce tailpipe emissions of carbon monoxide (CO) and certain hydrocarbons (HC), while contributing to lower lifecycle greenhouse-gas emissions, depending on the feedstock and production pathway.

Greater use of domestically produced ethanol can reduce dependence on imported crude oil.

The adoption of flex-fuel technology can contribute to India's vision of Atmanirbhar Bharat by promoting indigenous fuel production, enhancing energy security, and supporting the country's long-term goal of achieving net-zero emissions by 2070.

With this vision, the Government of India is encouraging the deployment of flex-fuel vehicles across the country through this regulation.